

Stablecoin Issuance and U.S. Treasury Bill Yields

A State-Dependent OLS Test, 2020M1-2026M4

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00 Title & Abstract

Stablecoin Issuance and U.S. Treasury Bill Yields: A State-Dependent OLS Test, 2020M1-2026M4

Dollar-pegged stablecoins, say, USDT and USDC, are now collectively holding close to US\$200 billion of short-term U.S. Treasuries, making their issuers among the largest buyers of T-bills while not representing a sovereign agency. Recent work at the BIS, the Federal Reserve, and in the academic literature has argued that stablecoin issuance drives down the short-end yields, with the effects intensified during periods when T-bills are scarce relative to the rest of the marketable debt stock. This paper tests both claims with monthly U.S. data from January 2020 to April 2026 ($T = 76$ months; effective $N = 74$ after lags), regressing the change in the 3-month T-bill yield on lagged growth in total stablecoin market cap, a T-bill scarcity dummy, their interaction, and standard controls (change of fed-funds-rate, VIX, post-UST-collapse dummy). The baseline specification yields the predicted negative signs on both stablecoin coefficients but is statistically insignificant. Under an alternative specification where volatility enters in monthly changes rather than levels, the state-dependent interaction term is large, negative, and marginally significant at the 10% level ($\hat{\beta}_3 = -2.37$, HAC $t = -1.89$, $p = 0.06$) which means a yield decline of roughly 50 basis points per two-standard-deviation inflow during scarcity months. The results are consistent in sign, though substantially larger in magnitude, reflecting the noise and lower power of monthly data.

Keywords: stablecoins, Treasury bill yields, OLS, dummy variables, interaction terms, state-dependent effects.

01 Introduction

As of the first quarter of 2026, according to Tether’s own Q1 2026 reserve attestation, the single largest issuer of U.S. dollar-pegged stablecoins holds approximately one hundred and forty one billion dollars of U.S. Treasury bills (T-bills) which accounts for 1.8 percent of all outstanding bills, which is comparable to a mid-sized sovereign holder. Adding the second-largest stablecoin issuer, Circle, and all the other smaller dollar pegs raises the sector’s combined T-bill position above one hundred and fifty billion dollars. In just over half a decade, an unregulated cluster of private sector entities has become a quantitatively meaningful price-setter at the very front of the U.S. yield curve.

The economic mechanism is trivial. Stablecoin issuers, by design and increasingly by regulatory requirement, back each token with short-dated dollar-denominated reserves. When new stablecoins are issued, the proceeds flow into Treasury bills more or less mechanically; when stablecoins are redeemed, T-bills are sold. If demand for T-bills is inelastic at the front of the curve, as the preferred-habitat literature has argued (Vayanos and Vila, 2021; Greenwood and Vayanos, 2014), those flows should be visible in yields. Three recent papers report exactly this result. Using daily data from 2021 to 2025 and an instrumental-variables strategy, Ahmed and Aldasoro (2025) estimate that a two-standard-deviation inflow into stablecoins compresses the 3-month T-bill yield by 2.5 to 3.5 basis points within ten trading days, with limited spillover to longer tenors and asymmetric responses to outflows that are roughly two to three times larger. Jacewitz (2025) documents that the effect roughly doubles during periods when bills are deficient relative to the rest of the marketable debt stock. Ante, Saggu, and Fiedler (2025) report a larger Tether-specific point estimate.

This paper provides a transparent monthly-frequency ordinary least squares test of the state-dependent prediction which includes the times when T-bills are scarce and it being sparse. Where the BIS and arXiv papers identify the effect using daily data and instrumental variables, I rely on lagged stablecoin growth and a multivariate set of

macro controls to mitigate, though not eliminate, simultaneity bias. The point of the exercise is not to improve upon the identification strategies of published literature. Instead, it is to ask whether the headline pattern, a negative average effect on the 3-month T-bill yield, amplified during bill scarcity, survives at a frequency and with a toolkit accessible to a third-year undergrad, using only the multiple regression methods covered in the early chapters of Wooldridge (2020).

From 2020 m1 to 2026 m4, I estimate that the predicted negative effect of lagged stablecoin growth on the 3-month T-bill is not statistically significant in the baseline specification. However, when contemporaneous volatility enters as a monthly change rather than a level, the more internally consistent specification given the rest of the model, the state-dependent interaction term becomes large and marginally significant at the 10% level: during months of bill scarcity, a one-percentage point of increase in lagged stablecoin growth predicts a 2.4 basis point decline in the 3-month yield ($\hat{\beta}_3 = -2.37$, HAC $t = -1.89$, $p = 0.06$). The implied conditional effect of a two-standard deviation inflow during scarcity months is roughly 50 basis points, being broadly consistent in sign with Ahmed and Aldasoro (2025) and Jacewitz (2025) but larger, reflecting the lower power of monthly aggregation.

The remainder of the paper is organised as follows. Section 2 reviews the relevant literature and develops two testable hypotheses. Section 3 describes the data sources and the construction of the main variables. Section 4 sets out the regression model and the expected signs of each coefficient. Section 5 presents the estimation results. Finally, Section 6 concludes the whole paper.

02 Hypothesis development

02.0 Related literature

The literature relevant to this paper falls into two strands. At the foundational level, the work on preferred-habitat investors and inelastic Treasury demand. Vayanos and Vila (2021) co-developed the canonical preferred-habitat model, in which investors care about specific maturity segments and treat bonds of different tenors as imperfect substitutes; under their framework, demand or supply shocks confined to a particular segment of the curve move yields locally rather than getting smoothed away across all maturities. Simultaneously, Greenwood and Vayanos (2014) provided complementary empirical evidence that exogenous changes in the maturity structure of government debt forecast excess bond returns, consistent with bond markets that are not perfectly arbitrated across the yield curve. The mechanism this literature describes here is precisely what is required for stablecoin inflows to move yields measurably, that is, a demand shock concentrated at the very front of the curve.

A more recent, rapidly expanding empirical literature applies these front-end demand mechanisms directly to stablecoins. Ahmed and Aldasoro (2025) use daily data from 2021 to 2025 and an instrumental-variables strategy that exploits idiosyncratic cryptocurrency price variation to identify the causal effect of stablecoin flows on Treasury yields, with conventional financial market correlations being removed. Their findings were as follows: A two standard deviation inflow reduces the 3-month T-bill yield by 2.5 to 3.5 basis points within ten trading days. Outflows, however, raise yields by two to three times as much. And, the overall effect is concentrated at the end of the yield curve. Jacewitz (2025) argues that although the stablecoin industry now holds a substantial position in T-bills, much of that demand is reshuffled from elsewhere in the financial system, mainly bank deposits and money market funds. Hence, the net effect on aggregate Treasury demand is smaller than the gross holdings suggest. He also reports that the yield response roughly doubles in months when bills are scarce relative to the

rest of the marketable debt stock. Unlike the BIS paper, Ante, Saggu, and Fiedler (2025) focus purely on Tether (USDT) alone, discovering a larger elasticity, specifically, when Tether's share of total T-bills market goes up by one percentage point, the one month T-bills yield drops by 14 to 16 basis points. However, compared to the BIS paper, they didn't use an instrument variable to rule out the effect of exogenous variation, resulting in a less rigorous methodology.

My contribution here is a monthly-frequency replication of the state-dependent prediction, using only the multiple regression toolkit in the Wooldridge textbook. Hence this exercise is not methodological. Instead, my paper asks whether the headline pattern is detectable in publicly available monthly data without the instrumental variables machinery of the published work (Ahmed and Aldasoro, 2025). Another contribution of my work is to extend the sample through April 2026, capturing the post-Terra era recovery in stablecoin issuance more thoroughly than earlier work and the early implementation of the GENIUS Act.

02.1 Hypothesis 1: Stablecoin price impact on the short end

The first hypothesis is the average effect prediction of the preferred-habitat literature and the published stablecoin yield work:

H1: Controlling for the stance of monetary policy, prevailing market volatility, and the post Terra/UST regime, lagged growth in total stablecoin market capitalisation is negatively associated with the contemporaneous monthly change in the 3-month Treasury bill yield.

In Section 4 this corresponds to $\beta_1 < 0$. The hypothesis rests on two assumptions. On the demand side, demand for T-bills at the very short end of the curve is inelastic enough that a non-trivial private sector inflow moves the equilibrium yield. Regarding quantity, stablecoin issuers do in fact deploy most reserves into T-bills rather than into other safe assets, so issuance is a workable proxy for marginal T-bill demand. Neither is

airtight, they are also the assumptions on which the published estimates rest, so this paper inherits whatever weakness they carry.

Lagging the stablecoin variable by one month is a deliberate but partial fix to simultaneity. Since yields and stablecoin flows are jointly determined within a month, a contemporaneous specification would confound the two channels. However, although lagging cleans up the most obvious form of reverse causation, it does not deliver clean causal identification which requires an instrument which Ahmed and Aldasoro (2025) supply while this paper does not. The estimated coefficient should accordingly be read as a conditional correlation, suggestive of a causal effect rather than equivalent to one.

02.2 Hypothesis 2: State-dependence in bill scarcity

The second hypothesis tests the state-dependent prediction that motivates the interaction term in the regression:

H2: The price impact of stablecoin issuance on the 3-month T-bill yield is amplified during months when treasury bills are scarce relative to the rest of the marketable debt stock.

In the regression, H2 corresponds to $\beta_3 < 0$, where β_3 is the coefficient on the interaction between lagged stablecoin growth and the bill scarcity dummy. The mechanism is standard inelastic-markets logic: the price impact of a demand shock rises when supply elasticity falls. When Treasury runs down bill issuance, as during the Treasury General Account rebuild after the debt ceiling episode in 2023, the marginal supply curve is steeper, so the same dollar of stablecoin inflow pushes the equilibrium price by more. Both Ahmed and Aldasoro (2025) and Jacewitz (2025) record this state-dependence at daily and monthly frequencies. The empirical question here is whether the pattern can still be observed after monthly aggregation while removing the instrument.

A second prediction follows from the same logic and is reported alongside the baseline regression in Table 2: during scarcity months, the total stablecoin effect equals $\beta_1 + \beta_3$, and the joint sum should be significantly different from zero.

03 Data & variable description

03.1 Data sources

Drawn from three sources, I base my analysis on data at the monthly frequency from January 2020 to April 2026 which covers two Fed tightening/easing cycles, the Terra-Luna collapse in May 2022, and the banking stress incident. The sample contains 76 monthly observations, of which 74 enter the regression after two months are lost due to the construction of the lagged stablecoin variable. The reason why the window begins in January 2020 is because stablecoins' macroeconomic footprint before then is negligible.

I collect monthly market capitalization for Tether (USDT) and Circle (USDC) from DefiLlama and sum the two into a single stablecoin aggregate. My decision to restrict attestation to these two issuers is based on the fact that these two are fiat-reserved-backed which means their reserves are held mostly in cash and short term U.S. Treasury bills which means token issuance maps directly into T-bill demand. Also, crypto-collateralized stablecoins such as DAI are excluded because of their partially held backing in Treasuries, resulting in a weaker mechanical link between their issuance and T-bill demand. When it comes to the outstanding volume among all stablecoins, USDT and USDC are also by far the largest dollar-pegged stablecoins, together accounting for the large majority of the sector by market capitalization. The series is observed daily; I take the end-of-month value for each calendar month and compute the monthly percentage growth rate which enters the regression with a one-month lag. For validation, the aggregate figures were cross checked against the monthly reserve attestations published by Tether and Circle.

I collect monthly series of U.S. interest rates and market volatility from FRED, the database maintained by the Federal Reserve Bank of St. Louis. The dependent variable is constructed from the 3-month treasury bill secondary-market rate (FRED series TB3MS). I also obtain the effective federal funds rate (FEDFUNDS) and the CBOE

Volatility index (VIXCLS); the VIX series is observed daily and is aggregated to a monthly mean. The bill scarcity is constructed from the U.S. Treasury monthly statement of the Public Debt, obtained through the Treasury Fiscal Data API. For each month I compute the ratio of the marketable Treasury bills outstanding to total marketable Treasury debt. A month is classified as a scarcity month ($SCAR = 1$) when this ratio falls in the bottom quartile of its sample distribution, which is a conventional way to separate a low-supply state from the rest of the sample. Also, it classifies 17 of the 74 months as scarcity months, enough variation to estimate the interaction term while keeping the dummy meaningful. The table below reports the definition and source of each variable; Table 1 reports summary statistics.

Variable	Description	Source
DY3M _t	Dependent variable. Monthly change in the 3-month U.S. Treasury bill secondary-market rate, in basis points: $DY3M_t = 100 (TB3M_S_t - TB3MS_{t-1})$, where $TB3MS_t$ is the monthly-average annualized rate in percent.	FRED, series TB3MS
LagDSC _t	Lagged monthly growth rate of combined USDT + USDC market capitalization, in percent: $LagDSC_t = 100 (SC_{t-1} - SC_{t-2}) / SC_{t-2}$, where SC_t is the end-of-month sum of the two issuers' market caps in U.S. dollars. Lagged one month to mitigate simultaneity with DY3M _t .	DefiLlama
SCAR _t	Bill-scarcity dummy. Equals 1 if $bill_share_t$ falls in the lowest quartile of its in-sample distribution, 0 otherwise, where $bill_share_t = \text{marketable Treasury bills outstanding} / \text{total marketable Treasury debt outstanding}$, both end-of-month.	Treasury MSPD (constructed)
LagDSC _t × SCAR _t	Interaction term, the product of LagDSC _t and SCAR _t . Its coefficient measures the extra effect of lagged stablecoin growth on the yield change during bill-scarcity months.	Constructed
DFF _t	Monthly change in the effective federal funds rate, in basis points: $DFF_t = 100 (FEDFUNDS_t - FEDFUNDS_{t-1})$, where $FEDFUNDS_t$ is the monthly-average rate in percent. Controls for the direct pass-through of monetary policy to short-term yields.	FRED, series FEDFUNDS
VIX _t	CBOE Volatility Index, the monthly mean of daily closing values. Controls for the flight-to-quality channel. In the preferred specification it is replaced by its first difference, $\Delta VIX_t = VIX_t - VIX_{t-1}$.	FRED, series VIXCLS

Variable	Description	Source
POST_UST_t	Structural-break dummy. Equals 1 for every month from June 2022 onward (after the May 2022 collapse of TerraUSD), 0 before. Captures any regime shift in stablecoin composition and reserve quality following the collapse.	Constructed

Table 1: Summary Statistics.

Variable	N	Mean	Min	Max	
DY3M (bp)	74	2.824	23.362	-123.000	74.000
LagDSC (%)	74	6.047	11.016	-4.682	68.518
DFF (bp)	74	2.784	22.560	-93.000	70.000
VIX	74	21.029	7.051	12.667	57.737
bill_share	74	0.202	0.029	0.151	0.255
SCAR	74	0.230	0.424	0	1
POST_UST	74	0.635	0.485	0	1

Note: Monthly data, January 2020 to April 2026 (N = 74 after the loss of two months to the lag structure). DY3M and DFF are expressed in basis points; LagDSC in percent; bill_share as a fraction of total marketable debt; VIX as the monthly mean of the daily index. SCAR and POST_UST are 0/1 indicator variables.

Sources: DefiLlama; FRED, Federal Reserve Bank of St. Louis; U.S. Department of the Treasury, Monthly Statement of the Public Debt.

04 Regression model

I estimate the following linear model by ordinary least squares. Because the data are monthly time series, the standard OLS standard errors may understate the true sampling uncertainty due to serial correlation in the residuals. I therefore report Newey-West heteroskedasticity and autocorrelation consistent (HAC) standard errors as the preferred inference for the main results, following the standard recommendations in Wooldridge (2020, Chapter 12). Every variable enters the regression as a monthly change, a growth rate, or a 0/1 dummy rather than as a level, so the model is estimated on series that are stationary, which means a series has a constant mean, a constant variance, and it keeps wandering around that fixed mean instead of drifting off somewhere, or plausibly so, which avoids the spurious-regression problems that levels of persistent can create. The point estimates are identical to those produced under standard OLS; only the standard errors and the resulting t-statistics and p-values change. The full model is down below.

$$\Delta y_t^{3M} = \beta_0 + \beta_1 \Delta SC_{t-1} + \beta_2 SCAR_t + \beta_3 (\Delta SC_{t-1} \times SCAR_t) + \beta_4 \Delta FF_t + \beta_5 VIX_t + \beta_6 POST_UST_t + \varepsilon_t$$

And below is coefficient interpretation:

1. β_1 : Basis point change in the 3-month yield from a 1pp rise in stablecoin market cap during non-scarcity months. H1: $\beta_1 < 0$.
2. β_2 : Average yield change in scarcity v non scarcity months (level effect).
3. β_3 : Additional yield response per 1pp of stablecoin growth when bills are scarce. H2: $\beta_3 < 0$.
4. β_4 : Direct fed funds pass-through which should be roughly one-for-one (+1) for a fully integrated front end.
5. β_5 : Risk off/flight-to-quality channel which is expected to be negative.
6. β_6 : Captures the post Terra-Luna regime shift in stablecoin position and quality.

7. $\beta_1 + \beta_3$: Total stablecoin effect during scarcity months.

The expected signs of each coefficient are as follows:

Coefficient	Expected sign	Reasoning
β_1 (LagDSC)	–	Inelastic-demand price impact at the short end.
β_2 (SCAR)	ambiguous	Scarcity per se may push yields up or down depending on what drives it.
β_3 (interaction)	–	Larger price impact when supply is thinner (Ahmed and Aldasoro 2025; Jacewitz 2025).
β_4 (DFF)	+ (≈ 1)	Direct policy-rate pass-through.
β_5 (VIX)	–	Flight-to-quality pushes bill yields down in risk-off episodes.
β_6 (POST_UST)	ambiguous	Post Terra-Luna collapse, USDC/USDT shifted further toward T-bills, which could deepen the channel.

I estimate this model in two forms. In the “baseline” specification (reported by Table 2), market volatility enters as the level of the VIX. Reported by Table 3, in the “preferred” specification, it enters as the monthly change in the VIX. The change form which is chosen before estimation instead of in light of which specification produces a significant coefficient is more internally consistent, because the dependent variable and the other controls (the T-bill yield and the fed funds rate) all enter as monthly changes, and mixing one level regressor in among differenced ones can distort the other coefficients. I report both so the reader can see whether the result depends on this choice.

05 Findings & tables

This section reports the estimated results. Table 2 gives the baseline regression with Newey-West HAC standard errors. Table 3 gives the preferred functional-form specification, when market volatility enters as a monthly change. Table 4 is a robustness check that replaces the binary scarcity dummy with the continuous bill_share variable. Table 5 reports the correlation matrix among the regressors, and Table 6 marks the residual diagnosis for the baseline model. The five tables are collected together below, which all of them will be discussed in sections 5.1 to 5.3 in turn. Significance is marked with *, **, and *** at the 10%, 5%, and 1% levels.

Table 2: Baseline regression. Dependent variable: monthly change in the 3-month T-bill yield (bp), Jan 2020 to Apr 2026, N = 74. Newey-West standard errors.

Variable	Coef.	HAC s.e.	t	p
Intercept	1.953	5.002	0.39	0.698
LagDSC (b1)	-0.047	0.071	-0.66	0.510
SCAR (b2)	-1.655	7.364	-0.22	0.823
LagDSC x SCAR (b3)	-0.228	1.283	-0.18	0.860
DFF (b4)	0.920***	0.136	6.74	<.001
VIX (b5)	-0.005	0.230	-0.02	0.984
POST_UST (b6)	-1.471	2.514	-0.59	0.560
Adj. R-squared	0.749			
F-test: b1 = b3 = 0	0.07			0.929
F-test: b1 + b3 = 0	0.06			0.800

Table 3: Preferred specification, volatility in changes. Dependent variable: monthly change in the 3-month T-bill yield (bp), N = 74. Newey-West HAC standard errors.

Variable	Coef.	HAC s.e.	t	p
Intercept	2.042	2.091	0.98	0.332
LagDSC (b1)	-0.076	0.060	-1.26	0.214
SCAR (b2)	17.002***	6.011	2.83	0.006
LagDSC x SCAR (b3)	-2.375*	1.255	-1.89	0.063
DFF (b4)	0.437***	0.103	4.23	<.001
dVIX (b5)	-0.011	0.308	-0.04	0.971
POST_UST (b6)	-4.164*	2.331	-1.79	0.079
Adj. R-squared	0.774			

Table 4: Robustness check, continuous bill_share interaction. Dependent variable: monthly change in the 3-month T-bill yield (bp), N = 74. Newey-West HAC standard errors.

Variable	Coef.	HAC s.e.	t	p
Intercept	1.254	16.713	0.08	0.940
LagDSC (b1)	0.610	1.390	0.44	0.662
bill_share (b2)	-2.892	89.601	-0.03	0.974
LagDSC × bill_share (b3)	-2.704	6.183	-0.44	0.663
DFF (b4)	0.914***	0.111	8.23	<.001
VIX (b5)	-0.001	0.242	0.00	0.996
POST_UST (b6)	-0.607	2.508	-0.24	0.810
Total R-squared	0.770			

Note: bill_share replaces the binary SCAR dummy and enters both on its own and interacted with lagged stablecoin growth. The standard error on bill_share (about 90) signals severe collinearity with the interaction term.

Table 5: Pearson correlation matrix among regressors (N = 74; N = 73 for pairs involving the change in VIX, which loses one observation to its own lag).

	VIX	Δ VIX	SCAR	LagDSC	LagDSC x SCAR
VIX	1.000	-0.060	0.355	0.159	-0.077
ΔVIX	-0.060	1.000	0.025	-0.203	0.123
SCAR	0.355	0.025	1.000	-0.313	-0.059
LagDSC	0.159	-0.203	-0.313	1.000	0.162
LagDSC×SCAR	-0.077	0.123	-0.059	0.162	1.000

Note: the 0.355 correlation between SCAR and the VIX level explains why the interaction term is absorbed in the levels specification of Table 2 but stands out once volatility enters as a change (Table 3), which means SCAR is essentially uncorrelated with the change in VIX (0.025). Source: PROC CORR on the estimation sample.

Table 6: Residual diagnostics for the baseline regression (Table 2).

Statistic	Value
Number of residuals	74
Mean	0.000
Standard deviation	11.208
Skewness	0.391
Kurtosis	3.524
Shapiro-Wilk W (p-value)	0.892 (< 0.001)
Kolmogorov-Smirnov D (p-value)	0.173 (< 0.010)
Anderson-Darling A-Sq (p-value)	3.236 (< 0.005)

Note: Skewness near 0 and kurtosis near 3 indicate an approximately symmetric, near-mesokurtic residual distribution which the normality tests nonetheless reject which is driven by a small number of large outliers in months of sharp policy movement. Inference in the main text uses Newey-West HAC standard errors, which do not require normal residuals. Source: PROC UNIVARIATE on the baseline residuals.

05.1 Interpretation

Table 2 announces the baseline regression. Both stablecoin coefficients come out negative, as predicted: -0.047 on lagged stablecoin growth and -0.228 on the interaction with bill scarcity while neither is close to significant. The HAC t-statistics are -0.66 and -0.18, and the joint F-test of $\beta_1 = \beta_3 = 0$ cannot be rejected ($F = 0.07$, $p = 0.93$). On the baseline numbers alone, then, there is no detectable stablecoin effect on the 3-month Treasury yield. The one variable that is sharply estimated is the fed-funds change: its coefficient of 0.92 ($t = 6.74$) sits close to the one-for-one pass-through that the theory predicts, a reassuring sign that the model is picking up the right macro dynamics. Most of the adjusted R-square of 0.75 comes from that single variable.

Table 3 re-estimates the model with the monthly change in the VIX in place of its level. This is the cleaner choice on functional-form grounds, since the dependent vari-

able and the other macro controls are all expressed as changes. After the re-estimation, the picture shifts: the coefficient of the new interaction term is now -2.375, with a HAC standard error of 1.255, a t-statistic of -1.89, and a p-value of 0.063, so the state-dependent effect is marginally significant at the 10% level. The bill scarcity main effect rises to 17.0 and clears the 1% level. The interaction coefficient implies that, during scarcity months, each additional percentage point of lagged stablecoin growth is associated with a 2.4 basis point fall in the 3-month yield. A large inflow of about 22 percentage points (two standard deviations) above the mean, would therefore scale to a yield decline of roughly 50 basis points. However, since the estimation is imprecise, the figure is best read as a rough magnitude instead of a precise number. Overall, the sign matches Ahmed and Aldasoro (2025) and Jacewitz (2025) while the monthly evidence is simply weaker and noisier than their instrumented results and daily data.

So why does the interaction differ this much between the two specifications? The bill-scarcity dummy and the VIX level are positively correlated in the sample, at 0.36 (see Table 5): the VIX averaged 25.6 in scarcity months against 19.7 in other months, because the scarce-bill months happened to fall in the high-volatility COVID and early-tightening period. When the VIX enters in levels, it soaks up variation that the scarcity dummy and its interaction would otherwise pick up, which pulls the interaction coefficient toward zero. The same dummy is uncorrelated with the monthly change in the VIX (0.03, Table 5), so entering volatility in changes breaks that overlap and lets the interaction stand on its own. I report both specifications so the reader can judge for themselves, although I prefer the change specification because of its internally consistent functional form instead of a bigger number from it. In both versions, the fed-funds pass-through is positive and strongly significant, and the coexisting volatility term is negligible, both economically and statistically.

05.2 Robustness

As a further check, the binary bill-scarcity dummy was replaced with the continuous `bill_share` variable and its interaction with lagged stablecoin growth. The results are reported in Table 4 above. In that specification, every coefficient of interest is statistically insignificant, and the `bill_share` term carries a standard error of about 90 on a point estimate of -2.89, a symptom of severe collinearity among `bill_share`, lagged stablecoin growth, and their product. Therefore, the continuous specification is uninformative at this sample size, and the dummy specification is the one that identifies the state-dependent effect. The reason why I choose to report this pattern which the effect is detectable under the dummy specification but not the continuous one instead of to suppress it is because it shows that the finding is fragile to reasonable changes in how scarcity is measured, and it reinforces the cautious reading of the result given in section 05.1. The fed-fund change remains positive and strongly significant in this specification as well, which confirms that the loss of significance is specific to the stablecoin-scarcity channel and not a sign of a misspecified model.

05.3 Residual diagnostics

One of the classical assumptions behind the exact t and F tests of OLS is that the regression errors are normally distributed (Wooldridge, 2020, Assumption MLR.6, Chapter 4). It is therefore standard practice to inspect the estimated residuals for departures from normality. Table 6 declares the relevant measures for the baseline regression. The residuals have a mean of zero by construction and a standard deviation of 11.21 basis points. Their skewness, a measure of asymmetry that equals 0 for a perfectly symmetric distribution, is 0.39, which indicates only a slight rightward tilt. Their kurtosis which is a measure of tail thickness which for a normal distribution is 3, is 3.52, indicating tails marginally fatter than normal. Both values sit close to their normal benchmarks, so the residual distribution is approximately bell-shaped.

Table 6 notes three formal tests of normality as well: the Shapiro-Wilk, Kolmogorv-Smirnov, and Anderson-Darling tests. Reporting all three rather than relying on one guards against the blind spots of any single statistic. Each compares the empirical distribution of the residuals with a normal distribution of the same mean and variance, summarises the gap between the two in a single test statistic, and returns a p-value. A p-value below the chosen significance level rejects the hypothesis that the residuals are normal. The three differ only in how they measure a gap. The Shapiro-Wilk test, the most widely used of the three, is based on the correlation between the ordered residuals and the values that would be expected if they had been drawn from a normal distribution, and it performs well at the sample size used here. The Kolmogorv-Smirnov test instead takes the single largest vertical distance between the residuals' cumulative distribution and the normal one. The Anderson-Darling test refines the Kolmogorv-Smirnov idea by placing extra weight on the tails of the distribution, which makes it the most sensitive of the three to the kind of large outliers found in this sample. All three reject here at conventional levels (Shapiro-Wilk $W = 0.892$ ($p < 0.001$), Kolmogorov-Smirnov $D = 0.173$ ($p < 0.010$), and Anderson-Darling $A-Sq = 3.236$ ($p < 0.005$)).

The rejection occurred mostly because of a small number of large outliers in the months of sharpest monetary-policy movement, visible in the residual histogram and quantile-quantile plot in the companion SAS output, rather than by any systematic skew. This failure of normality does not invalidate the inference reported above. With 74 observations, the OLS estimator is approximately normally distributed by the large-sample argument of Wooldridge (2020, Chapter 5), whatever the shape of the error distribution, and the Newey-West HAC standard errors used throughout do not require normal errors in the first place. The non-normality is best read as one further reason to treat the marginal interaction result as suggestive rather than decisive.

06 Conclusion

This paper asked whether US stablecoin issuance compresses short-term Treasury yields, and whether that effect is amplified when Treasury bills are scarce relative to the rest of the marketable debt stock. Using monthly data from January 2020 to April 2026 and an OLS regression with a dummy-by-continuous interaction in the spirit of Chapter 6 and 7 of Wooldridge (2020), I estimated the relationship with Newey-West HAC standard errors and a robustness check.

The evidence is suggestive but not conclusive. In the baseline specification, the stablecoin coefficients carry the predicted negative signs but are statistically insignificant, and the joint test of no stablecoin effect cannot be rejected. Under the functionally more consistent specification in which volatility enters as a monthly change, the state-dependent interaction term is negative and marginally significant at the 10% level, with a point estimate implying a sizeable yield decline during scarcity months. The sign of the effect matches the daily-data, instrumented findings of Ahmed and Aldasoro (2025) and Jacewitz (2025), but the monthly-frequency result is weaker, sensitive to the volatility specification, and does not survive replacing the scarcity dummy with a continuous measure.

Three limitations qualify the result. The sample being short, with only 74 months after the loss of two observations to the lag structure, the precision of any state-dependent coefficient is limited. Simultaneity bias being only mitigated, not eliminated, by lagging the issuance variable, a proper instrumental-variable approach such as the redemption-driven instrument of Ahmed and Aldasoro (2025) would be the natural next step. The bill-scarcity dummy only serving as a coarse proxy, a market-based scarcity measure such as the bill-OIS spread would sharpen identification. Subject to these limitations, the monthly evidence is consistent with the view that stablecoin issuance now exerts a detectable, state-dependent influence on the very front of the US yield curve, despite the current research design failing to establish that influence as causal.

07 Reference

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- Ante, L., Saggu, A., & Fiedler, I. (2025). The stablecoin discount: Evidence of Tether's U.S. Treasury bill market share in lowering yields. arXiv:2505.12413. <https://arxiv.org/abs/2505.12413>
- Greenwood, R., & Vayanos, D. (2014). Bond supply and excess bond returns. *Review of Financial Studies*, 27(3), 663-713.
- Jacewitz, S. (2025). Stablecoins could increase Treasury demand, but only by reducing demand for other assets. Federal Reserve Bank of Kansas City Economic Bulletin, August 8, 2025.
- Vayanos, D., & Vila, J.-L. (2021). A preferred-habitat model of the term structure of interest rates. *Econometrica*, 89(1), 77-112.
- Wooldridge, J. M. (2020). *Introductory Econometrics: A Modern Approach* (7th ed.). Cengage Learning.

08 SAS code

```
%let LIBPATH = C:\Users\User\OneDrive\Desktop\Econometrics\report;
```

```
proc import out=work.panel  
  datafile="&LIBPATH.\Stablecoin_data_2020_2026.csv"  
  dbms=csv replace;  
  getnames=yes;  
run;
```

```
proc sort data=work.panel; by date; run;
```

```
data work.pre;  
  set work.panel;  
  by date;  
  LagMcap = lag(mcap_bn);  
  LagLagMcap = lag2(mcap_bn);  
  DSC = 100 * (mcap_bn - LagMcap) / LagMcap;  
  LagDSC = 100 * (LagMcap - LagLagMcap) / LagLagMcap;  
  LagTB = lag(TB3MS);  
  DY3M = 100 * (TB3MS - LagTB);  
  LagFF = lag(FEDFUNDS);  
  DFF = 100 * (FEDFUNDS - LagFF);  
  POST_UST = (date >= '01JUN2022'd);  
run;
```

```
proc rank data=work.pre groups=4 out=work.ranked;  
  var bill_share;  
  ranks bs_q;  
run;
```

```
data work.analysis;  
  set work.ranked;  
  SCAR = (bs_q = 0);  
  DSC_x_SCAR = LagDSC * SCAR;  
  if cmiss(DY3M, LagDSC, SCAR, DFF, VIXCLS, POST_UST) = 0;  
run;
```

```
ods rtf file="&LIBPATH.\stablecoin_results.rtf" startpage=yes;
```

```
title "Table 1A. Descriptive statistics, continuous variables";
```

```
proc means data=work.analysis n mean stddev min p25 median p75 max maxdec=3;
  var DY3M LagDSC DFF VIXCLS bill_share;
run;
```

```
title "Table 1B. Frequency of dummies and the interaction";
proc freq data=work.analysis;
  tables SCAR POST_UST SCAR*POST_UST / nocol norow nopercent;
run;
title;
```

```
title "Table 2. OLS regression of monthly change in 3-month T-bill yield";
proc reg data=work.analysis plots=none;
  model DY3M = LagDSC SCAR DSC_x_SCAR DFF VIXCLS POST_UST / vif;
  output out=work.resid r=uhat p=yhat;
  h_no_stable: test LagDSC = 0, DSC_x_SCAR = 0;
  h_no_interaction: test DSC_x_SCAR = 0;
  h_scar_effect: test LagDSC + DSC_x_SCAR = 0;
run; quit;
```

```
title "Table 3. OLS with Newey-West HAC standard errors";
proc autoreg data=work.analysis plots=none;
  model DY3M = LagDSC SCAR DSC_x_SCAR DFF VIXCLS POST_UST
    / covest=neweywest dw=4 dwprob;
run;
title;
```

```
data work.rob;
  set work.analysis;
  LagVIX = lag(VIXCLS);
  DVIX = VIXCLS - LagVIX;
run;
```

```
title "Table 4. Robustness check 1: DVIX instead of VIX in levels (HAC SEs)";
proc autoreg data=work.rob plots=none;
  model DY3M = LagDSC SCAR DSC_x_SCAR DFF DVIX POST_UST /
  covest=neweywest;
run;
title;
```

```
data work.rob2;
```

```

    set work.analysis;
    DSC_x_bs = LagDSC * bill_share;
run;

title "Table 5. Robustness check 2: continuous bill_share interaction (HAC SEs)";
proc autoreg data=work.rob2 plots=none;
    model DY3M = LagDSC bill_share DSC_x_bs DFF VIXCLS POST_UST /
covest=neweywest;
run;
title;

title "Table 6. Correlation matrix among regressors";
proc corr data=work.rob nosimple noprob;
    var VIXCLS DVIX SCAR LagDSC DSC_x_SCAR;
run;
title;

title "Residual diagnostics";
proc univariate data=work.resid normal;
    var uhat;
    histogram uhat / normal;
    qqplot uhat / normal(mu=est sigma=est);
run;
title;

ods rtf close;

```